

who follows the method of Basu, Fernald, and Kimball (2006).¹⁴ Figure 7.5 shows that when there are periods of falling productivity, like the 1960s and 1970s, stock prices tend to fall. In the 1980s and 1990s (the computer revolution), productivity shocks are positive and stocks tend to appreciate. The correlation of the five-year moving averages of TFP shocks and stock returns is high, at 48%. So stocks are exposed to productivity risk; when productivity slows down, stock returns tend to be low.

Productivity risk is just one source of shocks that enter the new generation of *dynamic stochastic general equilibrium* (DSGE) macro models. This mouthful conveys the complexity of this class of models. In DSGE models, the economy is dynamic (as the name indicates) and the actions of agents (consumers, firms, central banks, and governments), technologies (how firms produce), and institutions or markets (the way that agents interact) cause economic variables to change. Asset prices are set from the complex interaction of all of these players and technologies. The DSGE models allow us to think about how shocks from these factor risks are transmitted across the economy. An important part of DSGE models are the actions of policy makers—government policy matters, as the financial crisis showed. Monetary policy and government shocks are important factors that influence asset prices and constitute their own sources of risk. We come back to some of these factors in the next few chapters. Current DSGE models nest both the real business cycle models pioneered by Kydland and Prescott, and they also include *new-Keynesian models*, where prices do not immediately adjust and inflation is *non-neutral*.

DSGE models describe business cycle fluctuations well, and we know asset returns vary over the business cycle. A benchmark model today is Smets and Wouters (2007), who specify seven shocks: productivity (as we have just discussed), investment, preferences, labor supply, inflation, government spending, and monetary policy. In chapters 8 and 9 covering equities and fixed income, respectively, we delve further into how some of these risks are priced.¹⁵

Demographic Risk

Another important risk for a very long-term investor is *demographic risk*. This can be interpreted as a shock to labor output, just as a productivity shock is a shock to firm production. A slow-moving variable, demography is a factor in economic *overlapping generations* (OLG) models. A given individual follows a life-cycle model, like the kind examined in chapter 5. Take, for example, an individual who progresses through youth, middle-age, and retirement. Labor income is earned and saved only during the young and middle-aged periods, and dis-saving occurs when retired. As any given age cohort progresses through the three stages, they join two other cohorts already alive who were born in previous generations. Thus, several

¹⁴ Available at <http://www.frbsf.org/csip/tfp.php>.

¹⁵ While most DSGE models do not study how these risks affect asset prices, the newest incarnations of these models, like Rudebusch and Swanson (2012), do.